

TABLE I | Two service capacities across ten CIM technologies

28 nm CMOS reference periphery · 128 × 128 INT8 · 16 KiB resident state

Technology / reference mode	Short budget			Typical / reference			Long budget		
	ρ [MB/s]	τ [MB/s]	RI* [-]	ρ [MB/s]	τ [MB/s]	RI* [-]	ρ [MB/s]	τ [MB/s]	RI* [-]
SRAM ACIM Binary charge-domain ACIM	83.12	7,273	0.01143	39.88	3,200	0.01246	16.62	1,600	0.01039
SRAM DCIM 16-term digital reduction	58.03	7,273	0.007980	49.81	3,200	0.01556	24.90	1,600	0.01556
2D NOR Binary sensing + digital reduction	4.901	0.07969	61.51	3.999	0.07969	50.18	3.569	0.009143	390.4
3D NAND SLC integration; sustained rewrite	0.07655	0.004571	16.75	0.07584	0.002327	32.59	0.07507	0.001347	55.73
RRAM 32-term ACIM; 16 write lanes	23.79	0.3310	71.88	12.49	0.2193	56.95	5.877	0.1080	54.40
MRAM Complementary MTJ; digital path	99.69	142.9	0.6978	49.81	84.21	0.5914	24.90	61.54	0.4047
PCM 8-row ACIM; diagonal updates	5.207	7.972	0.6532	2.500	6.874	0.3636	1.041	4.946	0.2106
HfO₂ FeRAM 1T1C digital; read-code reuse	68.82	307.7	0.2237	26.61	106.7	0.2495	12.88	53.33	0.2414
Gain-cell eDRAM Two-endpoint ACIM; refresh included	4.534	118.6	0.03824	1.804	66.79	0.02702	0.1403†	5.989†	0.02342†
3D FeFET 2026 vertical AND; digital path	22.71	11.27	2.016	11.10	9.249	1.200	5.551	6.838	0.8118

ρ : completed input-payload evaluation capacity. τ : resident-payload update capacity. RI* = ρ/τ : balance threshold.

Short / typical / long are paired engineering budgets, not confidence bounds. Typical means the selected reference design, not a statistical median.

† Gain-cell long: refresh stress only (3.93% service availability); excluded from its ordinary conditional range. All gain-cell rates include periodic refresh.

NOR / NAND use complete 16 KiB rewrites; other cases use the declared local update groups. The ten cases include volatile SRAM and gain-cell eDRAM.

Typical streaming and resident throughput

Ten CIM reference designs · $RI = \rho/\tau$

